

IoT-Based Smart Water Quality Monitoring System

¹N Sreekanth, ²N Ramani Mrudula, ³M Sruthi, ⁴P Manasa

¹Department of CSE, Associate Professor

^{2,3,4}Department of CSE, Students

BVRIT HYDERABAD College of Engineering For Women, Hyderabad, Telangana, INDIA.

Corresponding Author: Nara Sreekanth

ABSTRACT

Water quality management is a critical concern in maintaining public health and environmental sustainability. This project focuses on developing an IoT-based smart water quality management system designed to provide real-time monitoring and reporting of water quality metrics. The system integrates various sensors to measure parameters such as pH, turbidity, temperature, and humidity. Using the latest IoT technologies, data from these sensors is collected, processed, and transmitted to a centralized platform where it is analyzed and visualized. This predicts potential water quality issues and provides recommendations to improve the quality of the water. The proposed system aims to offer an affordable and scalable solution for continuous water quality monitoring, ensuring timely intervention and effective management of water resources.

Keywords: Water quality metrics, pH sensor, Turbidity sensor, Temperature sensor, Humidity sensor.

I. INTRODUCTION

Maintaining water quality is essential for sustainability and human health, but traditional monitoring methods are time-consuming, labor-intensive, and lack real-time capabilities.

An IoT-based Smart Water Quality Monitoring System addresses these issues by using sensors to measure pH, turbidity, temperature, and dissolved oxygen. Data is processed by a microcontroller (e.g., Arduino Mega or NodeMCU) and sent to the cloud via an ESP8266 Wi-Fi module for storage and visualization on platforms like ThingSpeak. Authorized users can access this data remotely for real-time monitoring and decision-making.

This system enhances scalability, precision, and cost-effectiveness, benefiting water treatment, agriculture, and industrial operations. By automating monitoring, it overcomes traditional limitations, providing real-time alerts, historical analysis, and user-friendly interfaces to ensure water safety in both urban and rural areas.

1.1 Objectives:

- A. Versatile System Design: Deployable in various contexts, IoT-based water quality monitoring systems integrate modern technology to enhance operational efficiency, regulate water purity, and address global water challenges.
- B. Predictive Analytics: Leverages real-time data and alerts to foresee contamination incidents, enabling preventive interventions and timely user actions.
- C. Customizable Sensors: Allows users to add or reduce sensors to meet specific water quality monitoring needs, ensuring adaptability to diverse requirements.
- D. Real-Time Recommendations: Provides visualizations of water changes and suggests modifications based on utility, aiding informed decision-making.
- E. Accessibility and Scalability: Caters to diverse user needs with scalable sensor integration, from basic setups to advanced configurations.
- F. Time and Effort Efficiency: Automates water testing and recommendations, saving time while increasing productivity and precision.

1.2 Existing Work:

An IoT-based real-time water quality monitoring system emphasizes cost-effectiveness and portability. Using NodeMCU for data acquisition and Wi-Fi-based platforms like ThingSpeak, it monitors pH, turbidity, temperature, and water level. Tested on various water samples, the system offers a compact, user-friendly design with real-time monitoring and alerts. While ideal for small-scale applications due to its affordability and low maintenance, its limited parameter range and lack of advanced analytics or machine learning may hinder scalability and broader use.

1.3 Proposed Work:

The water quality monitoring system is customizable, allowing users to add or remove sensors as needed, emphasizing simplicity and accessibility. Key features include:

- Real-Time Recommendations: Provides instant advice on modifying water for specific uses.
- Real-Time Preview: Displays water safety levels instantly, enabling users to make changes before use.
- Responsive Design: Ensures recommendations are clear, easy to understand, and user-friendly.

By overcoming limitations of existing tools, this cost-effective solution minimizes reliance on expensive products, making water testing accessible and convenient for all users.

II. LITERATURE REVIEW

2.1. Related Work:

[1] The paper *Low-Cost Flow Sensors: Making Smart Water Monitoring Technology Affordable* by Jessica Fell, Justin Pead, and Kevin Winter, introduces an affordable IoT-based solution for real-time water flow monitoring in developing regions. Using ultrasonic sensors and GSM-enabled devices, the \$46 system measures water levels via crump weirs and transmits data to a centralized server. It provides continuous insights into flow, rainfall, and pollutants, aiding drainage and wastewater planning in informal settlements. Benefits include cost-effectiveness, modularity, and real-time data, while challenges involve GSM reliance, power use, and theft risks. Tests on South Africa's Stiebeul River highlight its potential to improve water management in resource-limited areas.

[2] The paper *Behavioural Analysis of Water Consumption Using IoT-Based Smart Retrofit Meter* by Ayush Kumar Lall et al., presents a cost-effective IoT solution for monitoring water consumption by retrofitting analog meters with smart devices. Data from IIIT Hyderabad over four months revealed trends analyzed using deep learning (ResNet-18) for digit recognition and error correction. The system offers affordable, high-precision, real-time data for sustainable water management, though challenges include environmental interference, network outages, and maintenance.

[3] The paper *IoT-Enabled Water Monitoring in Smart Cities With Retrofit and Solar-Based Energy Harvesting* by Nilesh Bawankar et al. presents a cost-effective IoT solution for real-time water monitoring. It retrofits analog meters with IoT devices, uses edge computation with Random Forest (97.69% accuracy), and employs ThingSpeak for data visualization. Solar energy harvesting and power optimization extend battery life to 6 years. Tested on a 66-acre campus with 25 meters, the system is scalable, energy-efficient, and precise. Benefits include affordability, sustainability, and adaptability, while challenges involve network issues, reliance on sunlight, data security, and deployment complexity.

[4] The paper *"Quality Assessment and Monitoring of River Water Using IoT Infrastructure"* by Manish Kumar et al., published in the **IEEE Internet of Things Journal**, analyzes 15 months of data from the Ganga River and Sangam using the **Libelium Smart Water Kit** (98,851 samples). Sensors measured pH, dissolved oxygen, temperature, conductivity, and oxidation–reduction potential. Principal Component Analysis (PCA) and Factor Analysis were used to compute a region-adaptable *Water Quality Index (WQI)*. The system offers real-time, cost-effective monitoring and statistical feature prioritization but faces challenges like limited continuous monitoring, calibration needs, site-specific applicability, and network issues.

[5] The paper *"Renewable Energy Integration Into Cloud & IoT-Based Smart Agriculture"* by Et-Taïbi Bouali et al., published in **IEEE Access**, presents a cost-effective IoT-based Smart Agriculture system using Wireless Sensor and Actuator Networks (WSN/WAN). It employs fuzzy logic for irrigation scheduling and Node-RED for cloud-based data visualization. Powered by solar energy, the system, tested on a Smart Farm, monitors soil moisture, temperature, water levels, and energy metrics, reducing water usage by 71.8% and promoting sustainable agriculture. While scalable for off-grid and arid regions, challenges include reliance on solar power and limited generalizability due to single-site testing.

[6] The paper *"IoT-Based Water Management Systems: Survey and Future Research Direction"* by Shereen Ismail et al., published in **IEEE Access**, reviews 43 studies (2014–2022) on IoT-based water management in agriculture, industry, residential, and oilfield sectors. It highlights techniques like WSN, Cloud Computing, Big Data, ML, and AI, and proposes an optical water management system for oilfields using advanced sensors and multispectral imaging. While offering a comprehensive review and innovative solutions for water management, the paper lacks implementation of its proposed system, generalizes findings, and faces challenges with the cost and complexity of optical sensors.

[7] The paper *"IoT Based Smart Water Quality Monitoring: Recent Techniques, Trends, and Challenges for Domestic Applications"* by Farmanullah Jan et al., published in *Water* (2021), reviews advancements in IoT-based water quality monitoring. It highlights IoT integration, machine learning, cloud visualization, and

communication protocols like Wi-Fi, GSM, and LoRaWAN. Benefits include cost-effectiveness, real-time monitoring, scalability, and low energy use, while drawbacks include sensor limitations, biofouling, high maintenance costs, and data security concerns. The paper identifies system gaps, evaluates them empirically, and recommends strategies for efficient, scalable domestic IoT water monitoring.

[8] The paper "*IoT Based Smart Water Quality Monitoring System*" by Varsha Lakshmikantha et al., published in *Global Transitions Proceedings (2021)*, proposes a cost-effective IoT solution for real-time water quality monitoring. Using sensors for pH, turbidity, and conductivity, it integrates Arduino for data collection and Wi-Fi for cloud visualization. Tested on three samples, it is portable, user-friendly, and low-maintenance but limited in monitored parameters, lacking advanced analytics or machine learning, and may face challenges in diverse environments.

[9] The paper "*Smart Water Quality Monitoring System with Cost-Effective Using IoT*" by Sathish Pasika and Sai Teja Gandla, published in *Heliyon (2020)*, describes an IoT-based system for real-time water quality monitoring. Using data from Hyderabad's water supply, it measures pH, turbidity, water level, temperature, and humidity, transmitting results via NodeMCU to ThingSpeak for analysis. Benefits include cost-effectiveness, compactness, and real-time data, but its limited parameters and lack of advanced analytics restrict scalability and broader applicability.

[10] The paper "*A Temperature Compensated Smart Nitrate-Sensor for Agricultural Industry*" by Md. Eshrat E. Alahi et al., published in *IEEE Transactions on Industrial Electronics (2017)*, introduces a low-cost IoT-enabled nitrate sensor for real-time water quality monitoring. Using a planar interdigital capacitive sensor with parylene coating and electrochemical impedance spectroscopy, it detects nitrate levels (0.01–0.5 mg/L) with temperature compensation for accuracy. Data is transmitted via WiFi to a cloud server (ThingSpeak). Benefits include affordability (under \$100), high sensitivity, and IoT integration, but limitations include its range, ion interference, IoT dependency, and calibration needs. Validation with lab-prepared and stream samples confirmed reliability.

2.2. Research Gaps:

2.2.1. Focus on Specific Water Sources: The study primarily tests water samples from urban sources (e.g., Hyderabad Metropolitan water supply). The system's effectiveness in diverse water sources such as agricultural runoff, industrial effluents, or remote rural areas remains unexplored.

2.2.2. Energy Efficiency: The document does not address the system's energy requirements or propose solutions for optimizing power consumption, which is critical for remote and off-grid applications.

2.2.3. Long-term Sustainability: Issues such as sensor durability, energy efficiency, and system maintenance are often overlooked, limiting the sustainability of water monitoring solutions.

2.2.4. Public Awareness and Participation: Limited engagement and education of communities about water quality issues and how they can contribute to monitoring and conservation.

2.2.5. Dataset Limitations: Models often overfit due to small and homogeneous datasets. Research should explore larger, more diverse datasets for better generalization.

2.2.6. Preparing Hybrid Models: There is minimal research on hybrid AI models for enhanced segmentation. This is a promising area for future exploration.

2.2.7. Clinical Integration: Barriers to clinical adoption remain. Future studies should address usability and workflow integration in medical settings.

III Methodology

3.1 Proposed Model and Architecture

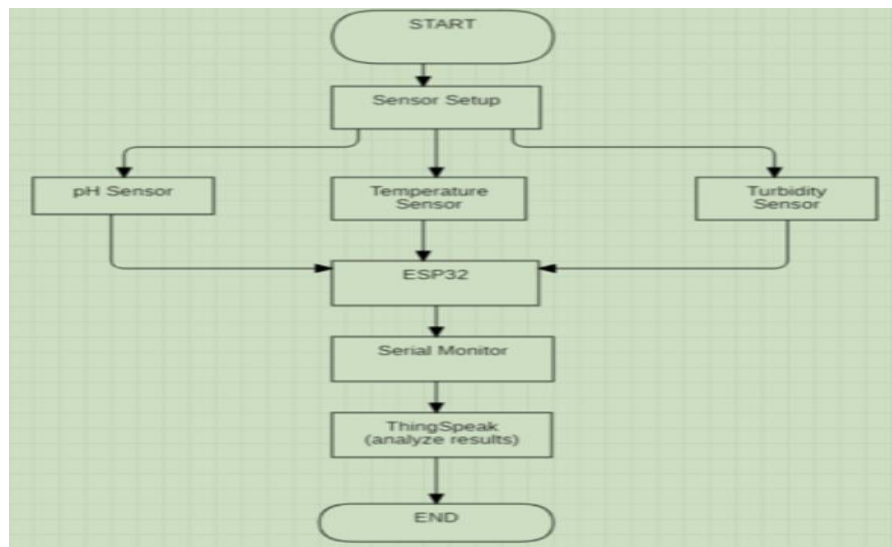


Fig: 3.1. Proposed Architecture for Water Quality Monitoring

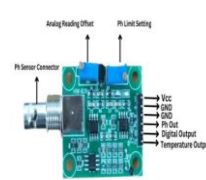
3.2. List of Components and Software used in this project.



ESP32 WROOM



pH Sensor



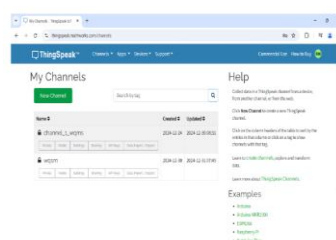
pH Sensor Connector



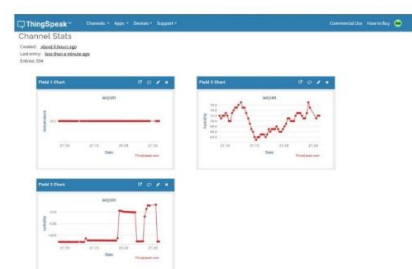
TS-300B
Turbidity Sensor



DHT11



ThingSpeak Channel



ThingSpeak Matlab Visualization

3.3. Methodology

Step 1: Start: Initiates the process.

Step 2: Sensor Setup: Set up the sensors required for water quality monitoring.

Step 3: Integration of Sensors:

- pH Meter: Measures the pH level using a pH sensor.
- Temperature Sensor: Monitors the water's temperature.
- Turbidity Sensor: Assesses the turbidity or clarity of the water.

Step 4: Connecting to ESP32: All sensor data is collected and processed using ESP32 microcontroller.

Step 5: ThingSpeak:

- Sensor data is transmitted to the ThingSpeak platform for real-time storage

and visualization.

Step 6: Analyze Results: Data from ThingSpeak is analyzed to determine the water quality.

Step 7: End: Concludes the process once water quality is assessed and regulated.

IV. DISCUSSION AND RESULTS

4.1 Discussion:

4.1.1 Strengths:

Ease of Use: The application reduces the complexity of the system, making it accessible to both technical and non-technical users. By abstracting the technical complexity, the platform is approachable for students and professionals with better utility recommendation experience.

Customizability: Users can adjust key sensor elements such as dissolved oxygen sensor, chlorophyll sensor, thus allowing personalization.

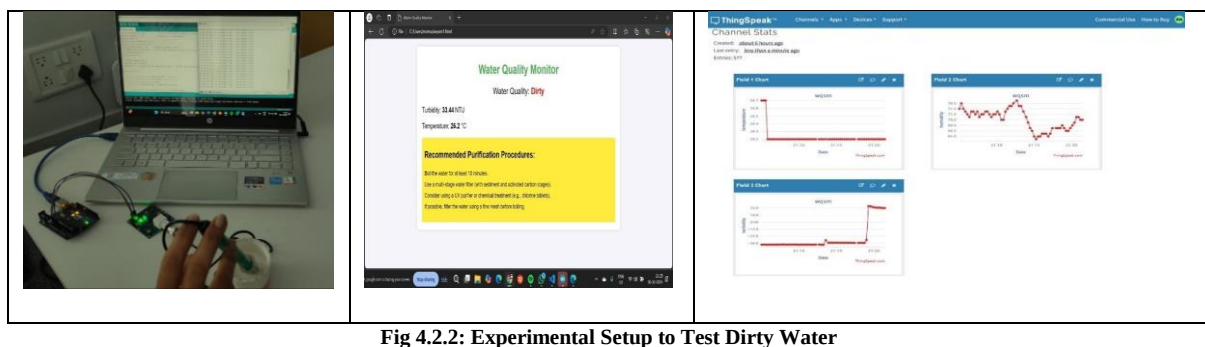
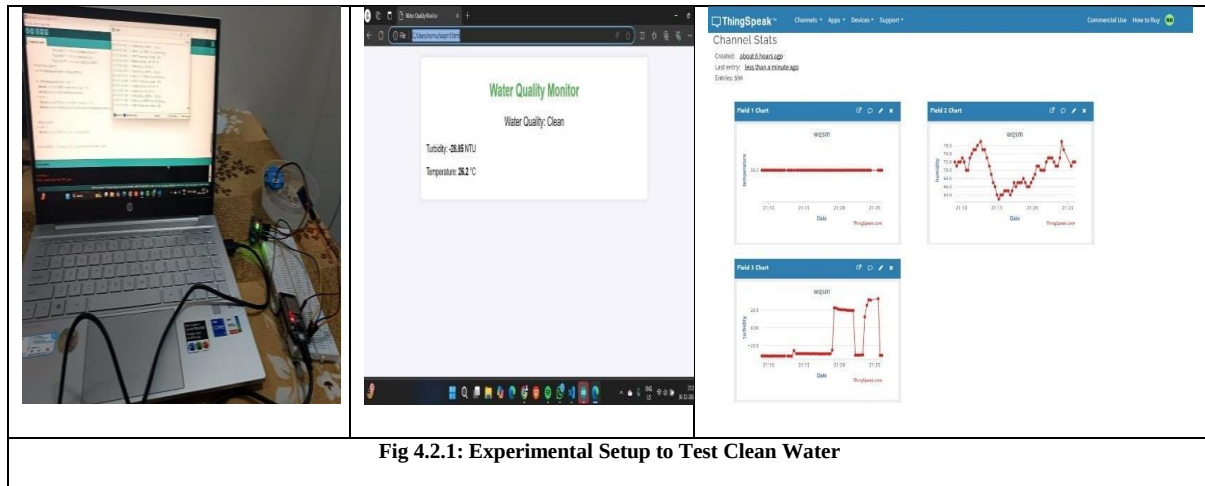
4.1.2 Limitations:

Basic Customization: While the customization options are functional, there is still room to improve on aspects such as more advanced design controls.

Scalability: The application is suitable for individual users or small-scale usage. However, as the number of users increases, further optimization might be required to ensure high performance.

Challenges: Sensor Management: Managing sensors when they are delicate to handle in a dynamic environment where users can add and remove sensor posed challenges. However, protection layers can be provided to the system which can be an effective solution for handling this complexity.

4.2 Results:



4.3 Comparison with Existing Solutions

- Compared to existing water quality monitoring systems, this solution offers greater flexibility in terms of customization of sensors and user interaction, especially in understanding the water state with real-time previews from Remote-Location.
- Existing tools often lack advanced customization features, whereas this system provides a higher degree of personalization and design flexibility.

V. CONCLUSION AND FUTURE SCOPE

5.1. Conclusion:

This IoT-based smart water quality management system presents a robust solution for enhancing water quality monitoring and management. By leveraging the power of IoT technologies, the system integrates sensors to collect real-time data on crucial parameters such as pH, turbidity, temperature, and humidity. This data is then transmitted to a centralized platform where it is analyzed, visualized, and utilized to predict potential water quality issues. This predicts potential water quality issues and provides recommendations to improve the water quality. The proposed system offers an affordable and scalable solution for continuous water quality monitoring, ensuring timely intervention and effective management of water resources. The system offers a cost-effective and scalable approach to continuous monitoring, empowering stakeholders with the necessary information to make informed decisions, implement timely interventions, and ultimately ensure the sustainable management of water resources.

5.2. Future Scope:

Integration with other systems:

- Integrate with existing water distribution systems and treatment plants to provide a more comprehensive view of water quality across the entire water cycle.
- Connect with weather forecasting systems to predict potential impacts of weather events on water quality.
- Integrate with citizen science platforms to involve local communities in water quality monitoring and reporting.

Enhanced User Interface and Data Visualization:

- Develop more user-friendly dashboards and visualizations to make data more accessible and understandable to a wider audience, including policymakers, water utilities, and the general public.
- Implement interactive maps to visualize water quality data spatially and identify areas of concern.

Expansion of Monitoring Parameters:

- Incorporate sensors for additional water quality parameters such as dissolved oxygen, conductivity, and heavy metals.
- Explore the use of biological sensors to detect the presence of harmful microorganisms.

Development of Mobile Applications:

- Develop mobile applications for citizens to report water quality issues, access real-time water quality information, and receive alerts.

REFERENCES

- [1]. Fell, Jessica, Justin Pead, and Kevin Winter. "Low-cost flow sensors: Making smart water monitoring technology affordable." IEEE Consumer Electronics Magazine 8.1 (2018): 72-77.
- [2]. Lall, A. Kumar, et al. "Behavioural Analysis of Water Consumption Using IoT-Based Smart Retrofit Meter." IEEE Access (2024).
- [3]. Bawankar, Nilesh, et al. "IoT-Enabled Water Monitoring in Smart Cities With Retrofit and Solar-Based Energy Harvesting." IEEE Access (2024).
- [4]. Kumar, Manish, et al. "Quality assessment and monitoring of river water using IoT infrastructure." IEEE Internet of Things Journal 10.12 (2023): 10280-10290.
- [5]. Bouali, Et-Taibi, et al. "Renewable energy integration into cloud IoT-based smart agriculture." IEEE Access 10 (2021): 1175-1191.
- [6]. Ismail, Shereen, et al. "IoT-based water management systems: survey and future research direction." IEEE Access 10 (2022): 35942-35952.
- [7]. Jan, Farmanullah, Nasro Min-Allah, and Dilek Düşteğör. "Iot based smart water quality monitoring: Recent techniques, trends and challenges for domestic applications." Water 13.13 (2021): 1729.
- [8]. Lakshmikantha, Varsha, et al. "IoT based smart water quality monitoring system." Global Transitions Proceedings 2.2 (2021): 181-186.
- [9]. Pasika, Sathish, and Sai Teja Gandla. "Smart water quality monitoring system with cost-effective using IoT." Heliyon 6.7 (2020).
- [10]. Alahi, Md Eshrat E., et al. "A temperature compensated smart nitrate-sensor for agricultural industry." IEEE Transactions on industrial electronics 64.9 (2017): 7333- 7341.